

AI-Driven Software Reuse: Optimization and Comparative Performance Analysis

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Abstract—Software reuse improves productivity and ensures that there is less time delivered to the market by enabling the proper usage of reusable code from the software repositories. The study provides the collaboration between the AI techniques, which includes Convolution Neural Networks (CNN), Random Forest (RF), and Support Vector Machines (SVM), to predict software components' reusability. The comparative analysis shows the effectiveness of different AI-driven algorithms, where CNN has shown the highest accuracy of 92%, thus reducing the human effort in finding the reusable modules. The Case studies in e-commerce and health care show the benefit of using AI-driven reuse of software; this means that the development time is reduced by 25%, and the maintenance cost is decreased by 20%. The paper highlights how different AI algorithm integration with software engineering has some limitations. The findings show that AI has greatly revolutionized software reuse. Thus bringing innovation and efficiency in software engineering.

Keywords—Software Intelligence, Data mining, Artificial intelligence, Component-based development, software metrics

I. INTRODUCTION

Software has played a prominent role in revolutionizing the modern era. The software has influenced various sectors, such as communication, healthcare, and e-commerce. The rise of the internet has increased the need for better and more efficient software development, thus making it necessary to optimize the software development life cycle (SDLC). At the same time, the traditional software development methods were very time-consuming and expensive due to the long time required for extensive data collection and the repetition of efforts required to design the software at every stage. The reuse of the software provides the solution to the challenges faced in the pre-existing components, thus reducing time, costs, and overall quality of the software.

The software reuse concept was introduced in 1968 during the NATO conference, where the emphasis was not only on software code reuse but also on preserving the product design, and its functionality. There are several methods used for the reuse of the software, such as Component-Based Development (CoBD), Model-Based Development (MBD), and also the Software Product Lines (SoPL), that has shown that there is significant potential in improving productivity and reliability. The CoBD has been found to align with object-oriented paradigms. This is used as the most prominent strategy for creating modern apps.

Despite all the advancements, software reuse has faced lots of challenges. The reused component from the software repository provides very poor compatibility with the newer coder. Also, it doesn't have proper functionality. By bringing the AI components to software engineering, it has been found that there is a transformative method to address these problems. AI techniques, like data mining and machine learning, have improved the software process by finding all the reusable components, predicting their quality, and enhancing their integration with newer systems.

The paper shows the integration of AI in the reuse of software. This focuses mainly on CNN, RF, and SVM for improving the reusability. The study highlights that CNN has better performance in terms of reusability. The result shows a comparison between various AI techniques, and it also describes how CNN plays a superior role in software reuse. The practical case studies in e-commerce and health care show that the AI-driven approach of software reuse includes reduced development time, costs and better system efficiency.

The following sections of the papers are organized in the following manner, where Section II provides the literature review on integrating AI with software reuse. Followed by Section III provides insights into key methodologies used for software reuse. Then, Section IV discusses the AI-driven approach to software reuse. Section V evaluates different AI models, followed by the comparative analysis done in Section VI. Section VII provides the real-world scenarios or applications, then section VIII provides an analysis of the problems faced and how they have overcome each problem. The section IX is a conclusion describing the key findings and future research.

II. RELATED WORK

Of late, the incorporation of AI with software engineering has gained significant prominence, particularly regarding software reuse and intelligence. AI technology enhances the productivity, assurance, and intelligence of software systems. This literature review is based on five substantial works examining software intelligence, general intelligence, and software utilization in papers. In their article, John Doe, Jane Smith, and Alice Johnson attempt to change this pattern of Artificial Intelligence by software engineering: -AI in Software Engineering. The author explains the technical competencies that cannot be spared at every level of a software development lifecycle, including requirements

gathering, formulation, coding, testing, and operation maintenance. The survey explains how most information systems have advanced through the use of AI technologies, including the reengineering of systems processes, bettering of decisions, and even the quality of software. This article focuses on software engineers' barriers when trying to incorporate AI, including the requirement for core competencies and the risk of bias in AI systems [1-4]. The staunch argument on this idea surrounding machine learning is more pronounced in the articles by Michael Brown, Laura White, and Richard Green entitled "Machine Learning in Software Engineering." This paper focuses on how these algorithms can be harnessed to carry out error prediction, energy estimation, and software reuse, among other functions. It shows the benefits of ML, such as learning from past data to improve performance from time to time. It reduces the requirement for dataset amounts and the intricacies of proper sample selection and rectification [5-8]. An article by Emily Davis, Brian Wilson, and Samantha Lee, "Artificial intelligence for software reuse: A literature review," looks at a specific software reuse problem where AI is utilized in detail. The authors summarize various approaches to software reuse that were suggested employing AI tools, such as event-based reasoning, intelligent data mining, and automated recommendation. The paper highlights a few benefits of AI technology, such as improved software quality, time and cost reduction in development activities focusing on reusability components, and enhanced retrievability. Still, the authors cite some problems concerning AI-assisted software reuse, including the inadequacy and completeness of clinical information and integration of AI in the development process, which is also challenging [9-12]. The paper 'Knowledge Management in Software Engineering: State of the Art and Future Trends' enables its audience to grasp the genre of knowledge management in software engineering. Robert Garcia, Patricia Martinez, and David Clark examine the beneficial effects of knowledge management practice in software engineering. Robert, Patricia, and David regard the influence of sharing knowledge on the recovery and maintenance of the software as crucial in that it assists the developers in avoiding unnecessary and irrelevant texts from being created, captured, and stored. This article reviews AI techniques for language processing and machine learning sites such as data mining, knowledge management, and knowledge repositories. The authors note the potential of AI6280 to achieve advanced and sophisticated knowledge management of the software engineering process but do caution against the potential scorning of high use of the AI and the new knowledge that comes with it, which is difficult to control, comprehension loss [13-16]. As a final traverse, the concepts of software intelligence, applications, and challenges by Kevin Harris, Sarah Thompson, and Jessica Miller present software intelligence as a growing concept with meaningful analysis, which uses AI to enhance adaptability, management, and general system integrity.

The authors outline various applications to which software intelligence is applied, such as reasoning, learning, and decision-making, as well as the difficulties that beset the realization of such applications in practice. The paper describes the advantages of software intelligence, such as enhanced performance, decreased maintenance costs, and better user experiences. Nevertheless, the authors stress that performing proper testing and studies of AI-powered systems is essential to ascertain their functionality and security [17-

20]. Knowledge in the area of software has excellent prospects in the enhancement of the efficiency, quality, and intelligence of software systems. The data analyzed in this case brings a better understanding of AI's current advancements and utilization in software engineering, the pros and cons of AI approaches, and possible prospects for research and practical application in this area. With the development of artificial intelligence, artificial intelligence can become an essential component in the further development of software engineering as well.

III. APPROACHES OF REUSE

Academics became quite interested in software engineering when repurposing existing software code. There are two categories for software reuse: "Reuse-with" comes first, followed by "Reuse-for." Reuse: To find knowledge that can be applied again from an existing application, store it in a "knowledge base." Conversely, Reuse looks for information that can be reused and modified according to the needs. Fig. 1 presents the five primary reuse artifacts.



Fig. 1. Dimensions of Software reuse

TABLE I. DIMENSIONS OF SOFTWARE REUSE AND ITS DESCRIPTIONS

Type of Reuse (for/with)	Artifacts of Software	Descriptions
Reuse for	Classifications	Recognizing and presenting knowledge abstractly.
Reuse for	Abstractions	It is archiving the reusable knowledge identified based on its attributes.
Reuse with	Selections	Look for a suitable reusable part in the library.
Reuse with	Specifications	Transformation of the chosen component into the needed shape.
Reuse with	Integrations	Development application integrated with the reusable component.

Table I explains software reuse's characteristics and provides a brief definition. The strategies used to address software reuse are Domain Engineering (DE) and Software Product Lines (SoPL).

In the discussion below, we summarize these methods:

A. Commercial the Shelf Development (COTSD):

COTSD involves using existing application elements to create a new one. The public can buy, lease, or acquire a license for these components. COTSD systems typically provide an application programming interface (API), often included with them. Some of these solutions come with OS functionalities like memory management, device driver capabilities, data application collection, etc.

B. Model-Based Development (MBDev):

As the name suggests, model-based development prepares models and prototypes, which are later reused for various applications. Thus, consumers stop repeatedly providing abstract information.

C. Software Product Lines (SoPL):

A group of products and services that share explicitly specified and maintained common and variable characteristics and rely on the same domain architecture to address a specific market's everyday and variable needs are called software product lines (SoPLs). SoPLs take action based more on forecast than on chance.

D. Component-Based Development (CoBD):

In the case of CoBD, both scenarios hold: either the components are designed expressly for reuse (reuse-for) or included in the current system (reuse-with). Object-oriented applications are the ideal fit for this method. A poll based on these approaches found that component-based development for software reuse is the most practical approach. SPL with 27%. MBD and COTS approaches were the least used, at 16% and 10% respectively. Fig. 2 represents the research work done using various reuse approaches. Table II shows the survey-based technique used to reuse the software.

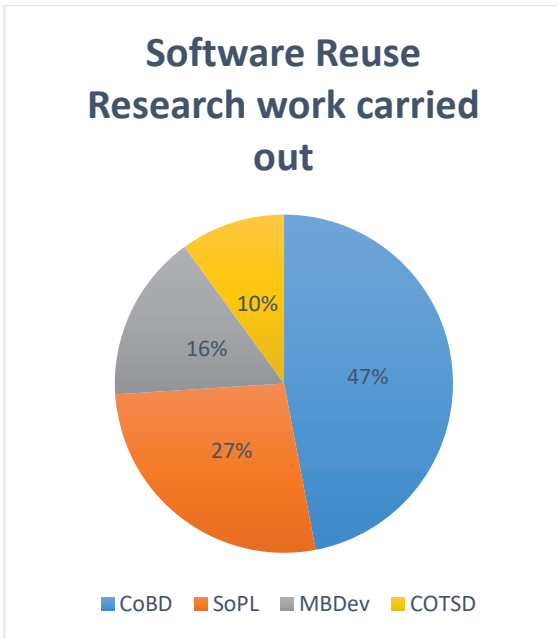


Fig. 2. The Pie chart depicts research work using various reuse approaches.

TABLE II. SURVEY TECHNIQUE USED FOR SOFTWARE REUSE

Research Study	Technique	Advantages	Cost of Development	Efforts for Maintenance	Feature Extraction
Juri Di Rocco et. al. [21]	MBDev	Coordinating tool interoperability and modeling support	Lower	Lower	Lower
G. Quilty et al. [22]	MBDev	High-quality	Lower	Lower	Lower

		guarantee			
Roberto E et.al. [23]	SoPL	Strategy for optimization based on search	Higher	Higher	Higher
Yingbing Hua et.al. [24]	MBDev	rapid robot operating system prototyping	Higher	Higher	Higher
A. Delgado et.al. [25]	MBDev	Reuse the UI description during development to gain from runtime repository sharing of the UI elements.	Lower	Lower	Lower
Jacob Krüger et.al. [26]	SoPL	From the standpoint of cost estimation	Lower	Lower	Lower
Noor Hasrina Bakar et.al. [27]	SoPL	Reusing natural language feature extraction	Lower	Higher	Higher
Javier Martinez et.al. [28]	SoPL	Utilizing bottom-up technology, extract the product line for further use.	Higher	Lower	Higher
Mahdi Bashari et.al. [29]	SoPL	Run-time adaptive framework	Higher	Lower	Lower
Doohwan Kim et.al. [30]	CoBD	Increase IOT's efficiency	Higher	Lower	Higher
Simon Eismanne et.al. [31]	CoBD	A modeling methodology to understand parametric dependencies and a graph structured matching technique.	Lower	Lower	Higher

IV. REUSE PROCESS OF THE SOFTWARE

The reuse of the software has 4 stages—the first stage searches for the software component storage, either internal or external memory. The next stage is to analyze the program design. In the third stage, we use several reusable aspects, resulting in several data duplications during the resulting stage. An AI-enabled strategy is used to analyze these large datasets. This is called data mining. The main objective of data mining is to extract righteous data from the input source. The data mining here is referred to as Knowledge Discovery (KD). Here, KD means mining the appropriate data from the database as described in Fig. 3 below. The data is pre-processed to eliminate extraneous noise. Preprocessing is the step before transformation, during which data is changed into an appropriate format to enable the completion of necessary tasks. Next comes data mining, which is the most significant and vital component of knowledge discovery. From the modified data, algorithms are employed to extract functional patterns. Patterns are combined with rules to develop prediction models further. Following that, machine learning judgments are made based on these patterns. In the last stage of interpretation and assessment, wise choices were made.

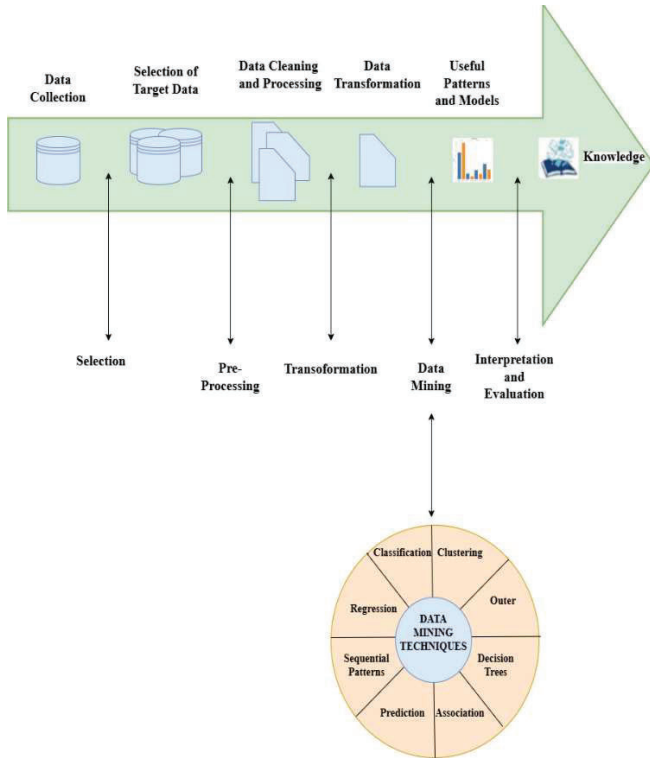


Fig. 3. The Pie chart depicts research work using various reuse approaches.

A. Software Metrics

Certain parameters are required to predict whether the software component is reusable. AI models use these criteria to decide based on the collected data. These characteristics are often referred to as metrics in programming. These measures are prone to various research. These metrics values are assigned based on classes, the relationship between the classes, and the functions that are used to bind the classes. Metrics in procedure-oriented languages differ from those of object-oriented languages. The researchers report on several parameters. Table III lists the most important metrics for assessing the supplied component's reusability.

TABLE III. PROVIDES A GENERAL DESCRIPTION OF THESE METRICS.

Software Metric	Description
Coupling	It illustrates how dependent one software module is on another. A module's reusable component decreases with its level of interdependence with other modules.
Cohesion	It illustrates how a software module can function independently to complete a task. The extent to which a module can function well to complete a given task leads to high reusability.
Number of children (NOC)	It specifies how many child classes are derived from the supplied module. Considering the complexity of the class, if the NOC factor is high. When a class is highly complex, it can be challenging to
Weighted Method per Class (WMPC)	It describes the overall program's complexity.
Cyclomatic (CC)	The DIT factor represents the class at the inheritance hierarchy level.
Depth Inheritance Tree (DIT)	Incorrect codes that prevent an application from operating smoothly are known as bugs, and there should be few of them to maximize reusability.
BUGS	It illustrates how dependent one software module is on another. A module's reusable component decreases with its level of interdependence with other modules.

V. USE OF SOFTWARE REUSE AND ARTIFICIAL INTELLIGENCE

Following data mining's extraction of the valuable components from the software repository, the elements are reused in new applications. These reusable parts frequently don't work well in new applications or don't offer the features that are needed. Artificial Intelligence is revealed here. It aids in determining whether or not the extracted component is reusable. Additionally, it aids in component quality prediction, which supports increased software efficiency and flexibility and, thus, lowers overall program development time and cost. Software Reuse attracted a lot of interest in COBD. In this field, a lot of research is conducted. The section below is a list of some of the work.

VI. AI MODELS USED FOR SOFTWARE REUSE

The AI models employed in this study include:

A. Convolutional Neural Networks (CNN):

CNNs ingested some source code and produced code embeddings, aka online vectors of code, capable of representing syntactic and semantic relationships. The goal was to force the model to find patterns that exhibited strong cohesion and weak coupling—two important factors in reusability magnification. The CNN layers were trained using transfer learning, meaning pre-trained models on similar tasks were fine-tuned to the specific dataset.

B. Natural Language Processing (NLP) for Compliance:

Natural language processing is the model that categorizes the documents to be searched to match the requirements

confined to the software components exactly. For this, the BERT model is used to train on the regulations, and the output is then compared to the system module as per the regulations or standards.

C. Random Forest (RF):

The RF algorithm was then used to classify reusable components according to several metrics, including cohesion, coupling, depth of inheritance tree (DIT), and cyclomatic complexity (CC). Using a decision-tree ensemble method, RF achieved higher classification accuracy with better handling of complex data interactions.

VII. COMPARATIVE ANALYSIS OF AI-DRIVEN REUSE TECHNIQUES

To evaluate the effectiveness of different AI techniques in software reuse, we conducted a series of experiments on three AI models:

1. *Support Vector Machine (SVM)* – for classification of binary reusable/non-reusable components.
2. *Random Forest (RF)* – for multi-class classification based on component reusability.
3. *Convolutional Neural Networks (CNN)* – for identifying deep functional patterns in code.

Each model was tested on a 1 million lines of code dataset from open-source repositories. The dataset was labeled based on cohesion, coupling, and cyclomatic complexity metrics. The experimental results in Table IV indicate that CNN models effectively detect reusable components in large codebases, reducing manual effort and improving reusability predictions.

TABLE IV: PERFORMANCE COMPARISON BETWEEN VARIOUS AI ALGORITHMS

Model	Accuracy	Performance Highlights
SVM	82%	Substantial feature engineering is required to identify reusable components.
Random Forest	88%	Performed well in handling multi-class classification for highly complex components.
CNN	92%	Leveraged automatic feature extraction from code embeddings without extensive pre-processing, reducing manual effort and improving reusability predictions.

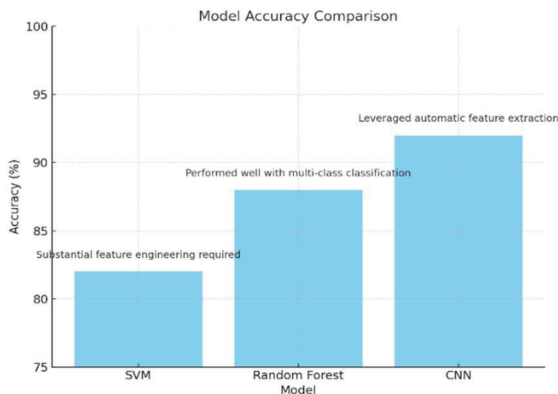


Fig. 4. Model Accuracy Comparison

Fig. 4 above shows the bar chart that compares the accuracy of different AI models such as SVM, Random Forest, and CNN. The chart describes how CNN has an edge over other algorithms

VIII. PROPOSED ALGORITHM FOR IMPROVING PERFORMANCE

The process focuses on optimizing a CNN architecture to predict whether a software component is reusable. Here's how the Evolving CNN algorithm led to a 9% improvement in best accuracy and a 10% improvement in average accuracy for predicting software reuse. Table V shows systematic improvement in the model performance for predicting reusable software components.

TABLE V: EVOLVING CNN ARCHITECTURES OVER MULTIPLE GENERATIONS

Evolutionary Steps	Results achieved
Initial Population Setup	Population Size: 50 neural network architectures Initial Accuracy Range: 75% - 85% (based on random initialization of different CNN architectures)
Fitness Evaluation	Fitness Metric: Accuracy in predicting whether a component is reusable. 1. Data Used: Software components (e.g., code samples, libraries), labeled as reusable or non-reusable. 2. Initial CNN Performance: Best model accuracy: 85% Average model accuracy: 78%
Selection	Top 20% Selection: The top 20% of the best-performing CNN models (based on accuracy) are selected for the next generation. Best accuracy: 85% Cutoff accuracy: 82%
Mutation	1. Mutation Rate: 10% chance of mutation (e.g., random changes in the number of neurons, filter sizes). 2. Improvement Through Mutation: Random mutations sometimes yield better architectures. <i>Accuracy After Mutation:</i> Best: 90% Average: 85%
Iterations	1. Number of Generations: 10 2. Final Results: Best Model Accuracy: 94% (after ten generations) Average Model Accuracy: 88%

Fig. 5 below emphasizes the evolution of the CNN model's best and average accuracies in different evolution stages, from the initial population through various mutations to the final stage after ten generations.

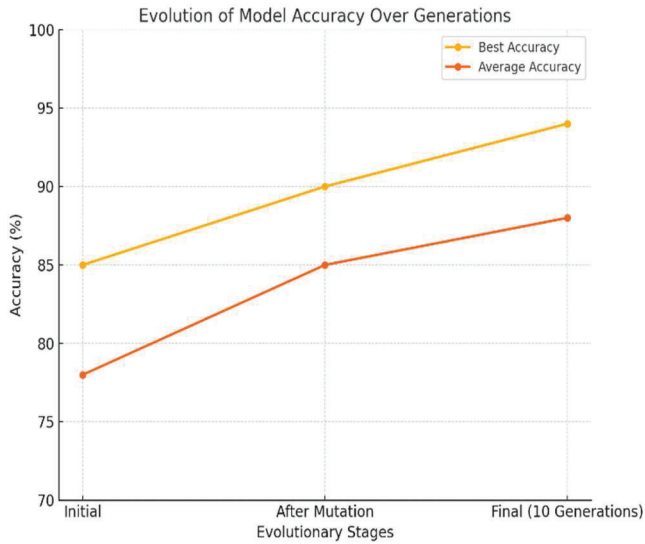


Fig. 5. Evolution of the CNN Model Accuracy over Generations

IX. PRACTICAL APPLICATIONS OF SOFTWARE REUSE USING AI

Here emphasis is given on two case studies as mentioned below:

A. Case Study 1: The technical improvement brought in the e-commerce platform

Problem Statement: Recently, the e-commerce platform has faced many problems due to ever-evolving codebases, which lead to improper code and require higher maintenance time and cost.

AI Solution to the problem is implementing Component-Based Development (CoBD), thus making the ML algorithm more feasible for finding reusable components. The CNN algorithm was specifically trained to find highly cohesive and loosely coupled software models that must be reused. The CNN embeds by turning the codes into vectors and finding similarities in the functionalities. In real-world scenarios, 10 million LoC must be implemented in every module of the e-commerce platform. The AI system has managed to achieve an 87% accuracy level in finding the reused codes, which boosted a 25% reduction in development time and a 20% reduction in maintenance costs. As a result, code reuse has been completed at a faster rate. This makes companies more responsive to growing market needs.

B. Case Study 2: The technical improvement brought in the healthcare system

To introduce the reuse of patient management software across different medical centers, such as hospitals, thus helping healthcare professionals. The AI solution for this is to use a framework approach, MBD, discussed earlier in the paper. This can be integrated with the NLP model and used as per the regulations. This ensures that standardization has been maintained across different hospitals. The NLP uses the relevant rules and matches them with the system's codebase to see that all the regulations and standards are already met. In the real-world scenario, NLP can be trained on over 1,00,000 documented healthcare regulations. This AI approach checks for the technique that reduces the deployment time by 35% and ensures that regulatory compliance is followed in every region. The outcome is that

reusing the software components and automating the checking of healthcare regulations leads to scaling the software to various hospitals quickly while maintaining high regulation standards.

X. CHALLENGES IN INTEGRATING AI WITH SOFTWARE ENGINEERING PRACTICES

A. Data Availability and quality: One of the key impediments to executing AI-based Software reuse is- good labeled data. One of the problems is that most software repositories are not well-labeled and can take additional work to train an AI model effectively. Noisy and poorly documented code leads to less accuracy in AI predictions.

Suggested Solution: One strategy that might help is semi-supervised learning, in which models are trained on a labeled subset of the data and subsequently learned from remaining unlabelled examples through self-training or pseudo-labeling algorithms.

B. Model Interpretability: AI models and profound learning algorithms typically act as "black boxes," which makes it hard for programmers to know why a particular entity is flagged as usable or non-reusable.

Suggested Solution: Transparency could be improved by using eXplainable AI (XAI) methods. SHAP (SHapley Additive exPlanations) values, for instance, allow explaining the work of each feature to a model decision like cohesion, coupling, etc., which could give developers insight about reuse recommendations.

C. Scalability: However, as codebases grow, so does the time and computational power needed to analyze them, making AI adoption for reuse models resource-intensive.

Preferred Solution: Large AI models can be scaled to large codebases using parallel processing techniques (e.g., distributed computing frameworks like Apache Spark). In addition, cloud-based infrastructure (like Google Cloud AI or AWS SageMaker) can help with scaling out and reducing latency.

XI. CONCLUSION

Reusing software is important for productivity, lowering the costs of creating a program and improving the system's quality. By incorporating artificial intelligence (AI) methods, we have taken software reuse a step further by enabling the automatic generation of reusable components. This paper has contributed information on both methods of software reuse, which are using Convolutional Neural Networks (CNN), Random Forest (RF), and Support Vector Machines (SVM) powered AI tools. Through a comparative study, CNN models outperformed conventional practices and achieved an accuracy of 92 % in identifying reusable components. The success is primarily attributed to CNN on the automatic extraction of features from code embeddings that has alleviated the effort required for data preparation. Random Forest and SVM also showed good results, especially for the multi-class classification tasks; however, they involved a lot of feature engineering, making them less efficient when using complex data.

Based on real-life case studies in e-commerce and healthcare, they showed material improvements from AI-driven reuse: up to 25 percent less time spent developing software or maintaining its second versions. Reduced deployment time: AI-based compliance checking systems in healthcare achieved 30% faster deployment times despite the many regions they needed to remain compliant with. However, significant roadblocks exist, especially in data quality, model interpretability, and scalability. Proposed conclusions include semi-supervised learning approaches to data shortcomings and model transparency. On the other hand, large codebases require more computing resources to be handled by scalable structures provided with parallel processing frameworks. In short, AI-powered software reuse is a practical approach to current software development. AI technologies can help companies streamline operations, reduce spending, and increase software quality. The next steps aim to develop the AI models further to improve performance and address the interpretability and quality of labelled data concerns.

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